

1 Physical Primitives of Mind: Modeling
2 Consciousness via Symmetries and Landauer
3 Erasure

4 Sebastian Bobadilla-Suarez
 `sebastian.bobadilla.s@gmail.com`

5 August 31, 2026

6 **Abstract**

7 We present a mathematical synthesis and formal Lean 4 verifi-
8 cation demonstrating how cognitive architectures can emerge as at-
9 tractor states from established physical primitives, without introduc-
10 ing new fundamental physics. Starting from the Poincaré group of
11 spacetime symmetries and finite phase space, we construct a frame-
12 work for predictive processing. Spontaneous symmetry breaking gen-
13 erates necessary topological boundaries, and finite phase space sub-
14 jects these boundaries to Landauer’s thermodynamic erasure. Agency
15 arises when systems evolve along paths of least action, minimizing en-
16 tropy production via internal structural resonance with external per-

17 turbations. Scaling this dynamic to macroscopic levels aligns with
18 emerging paradigms in neuroscience—specifically classical electrody-
19 namic fields (ephaptic coupling)—and Kuramoto phase synchroniza-
20 tion, culminating in a topological fixed point: a unified “Self.” We
21 state the claim narrowly, because the subject matter invites a wider
22 reading than the work supports: no new physics is derived here. Every
23 physical result the chain invokes is established—Picard–Lindelöf, Kib-
24 ble’s classification of topological defects, Landauer’s erasure bound,
25 Gibbs’ inequality, Barbalat’s lemma, Sakaguchi’s threshold, Banach’s
26 fixed point. What is new is that they are stated in one formal lan-
27 guage, their consistency mechanically checked, and the points where
28 the chain relies on stipulation rather than derivation located exactly.
29 The development declares no axioms and every structure carrying
30 physical content is inhabited by a witness; the synchronization step is
31 proved from a dynamics for initial data inside an explicit basin and
32 assumed outside it, and the identification of a global section of the
33 probability sheaf with the unity of experience is a stipulation, marked
34 as such throughout rather than absorbed into the derivation. The
35 framework is nonetheless falsifiable in one place that is its own and
36 no rival’s: recovery from sudden awakening should follow a delayed
37 sigmoid rather than an exponential, because the variable gating it is
38 astrocytic volume on a timescale of seconds to minutes rather than cor-
39 tical phase realignment on a timescale of milliseconds. As a corollary,
40 we show why rigid von Neumann architectures (e.g., GPUs) are ther-

41 modynamically disadvantaged or structurally constrained from reach-
42 ing these specific continuous thermodynamic attractors.

43 **1 Introduction**

44 The pursuit of a physical theory of consciousness typically involves search-
45 ing for the biological correlates of mind or treating subjective experience as
46 an abstract computational algorithm. This paper asks a more fundamental
47 question: If we assume only basic physical primitives—continuous symme-
48 tries (the Poincaré group), geometry, and thermodynamics—is the emergence
49 of a unified, perceiving entity a natural physical attractor?

50 We propose a framework where the structural properties of consciousness
51 are derived as physical attractor states. Establishing a deductive chain from
52 basic principles, we demonstrate how continuous symmetries break to form
53 boundaries, how those boundaries dissipate free energy as non-equilibrium
54 steady states, and how their evolution along paths of least action results
55 in physical resonance with their environment. This eventually scales into
56 the macroscopic electromagnetic unity proposed by emerging field theories
57 of mind. To avoid critiques of “math theatre,” we delineate between literal
58 physical mechanisms (e.g., thermodynamic erasure via Landauer’s principle,
59 boundaries from spontaneous symmetry breaking) and theoretical mappings,
60 utilizing Lean 4 formalization as our mathematical anchor. We adopt a per-
61 spective aligned with Russellian Monism: objective measurements of macro-

62 scopic electromagnetic fields constitute the third-person “outside,” while the
63 system’s internal physical deformation to reach structural resonance consti-
64 tutes the first-person intrinsic experience.

65 **What is new here, and what is not**

66 A framework paper about consciousness invites a particular suspicion, and it
67 is better answered at the start than at the end. **No new physics is derived**
68 **here.** Every physical result the chain invokes is established, and most of
69 it is old: Picard–Lindelöf for the well-posedness of the coupled dynamics
70 (Coddington & Levinson, 1955); Kibble’s classification of topological defects
71 by the homotopy of the vacuum manifold (Kibble, 1976); Landauer’s bound
72 on the heat cost of erasure (Landauer, 1961); Gibbs’ inequality for the relative
73 entropy (Cover & Thomas, 2006); Barbalat’s lemma for the asymptotics of a
74 dissipating trajectory (Barbalat, 1959); the threshold $K_c = 2D$ for the noisy
75 mean-field Kuramoto model (Sakaguchi, 1988); the bound of Still, Sivak,
76 Bell and Crooks relating dissipation to nonpredictive information (Still et
77 al., 2012); and Banach’s fixed-point theorem (Banach, 1922). A physicist
78 reading this paper will recognize all of it and would establish the point in
79 three paragraphs if we did not. Conceding it costs nothing.

80 What is new is of a different kind, and it is a smaller claim than a theory
81 of consciousness usually makes. These results have been stated in one formal
82 language, their mutual consistency has been mechanically checked, and the
83 places where the chain relies on stipulation rather than derivation have been

84 located exactly. The value of a machine-checked chain is not that any link is
85 novel; it is that no link can quietly borrow from another, that a hypothesis
86 cannot be smuggled in as a definition, and that the seams — where physics
87 stops and interpretation begins — are fixed in the source rather than nego-
88 tiable in the prose. Section 2.1 reports what the check cost: three of five
89 original axioms each proved **False**, from one shared root cause, and none of
90 the three had been caught by reading.

91 The framework is not thereby unfalsifiable, and one of its predictions is
92 its own rather than a rival's. Section 11 argues that recovery from sudden
93 awakening should follow a *delayed sigmoid* rather than an exponential, be-
94 cause the variable gating recovery is astrocytic volume and cerebrospinal-fluid
95 clearance, on a timescale of seconds to minutes, while the cortical rhythms
96 whose realignment is usually taken to constitute the recovery move in mil-
97 liseconds. That is a claim about functional form; it is fittable against data of
98 a kind already collected; and a purely electrical account of the same recovery
99 predicts a different form.

100 **2 Formalization in Lean 4**

101 A core contribution of this work is the verification of our deductive chain
102 using the Lean 4 interactive theorem prover. This approach builds on recent
103 efforts to formalize consciousness and brain semantics, such as the 'OnlyOne'
104 system (Scherf, 2026) and sheaf-theoretic models of neural manifolds (Inoué,

2026). It is necessary to explicitly distinguish which claims have been math-
 ematically derived as theorems within the proof assistant, and which remain
 irreducible physical postulates. Table 1 maps the primary claims of this
 paper to their current formalization status.

Table 1: Mapping of physical claims to Lean 4 formalization status. The development declares **no axioms**: `#print axioms` on any result above reports only `propext`, `Classical.choice` and `Quot.sound`. Earlier drafts carried five axioms, three of them inconsistent; all five became class fields, so a postulate is an obligation on each instance rather than a global assertion (Section 2.1), and two of the five have since been eliminated entirely (Derivation 5). “Theorem” still does not mean “physically established”: the postulates live in hypotheses and instance obligations, and every such class is inhabited by a witness in `Examples.lean`, so no row is vacuous.

Conceptual Claim	Lean 4 Implementation	Status
Pointwise Vacuum Minimization	<code>pointwise_vacuum_of_global_min</code> (Phase1), potential energy defined as an integral; holds a.e., and a double-well witness proves the a.e. cannot be strengthened	Theorem
Landauer’s Principle	Derived from statistical mechanics (Phase3); the axiom-free reversibility core becomes a heat bound via the field <code>StatisticalMechanics.heat_eq</code>	Theorem + instance postulate
Symmetry Breaking implies Boundaries	Derived from pointwise potential minimization (Phase1); witnessed on a double well with a $\mathbb{Z}_2$ symmetry whose minimisation hypothesis is proved, not assumed	Theorem
Inevitability of Boundaries (π_0)	The domain-wall case of the Kibble mechanism (Phase1 §3): on a connected substrate, a continuous field taking values in two different connected components of the vacuum manifold must leave the vacuum somewhere (<code>exists_notMem_of_connectedComponentIn</code>). Witnessed on the double well, whose vacuum manifold is computed to be exactly $\{-1, 1\}$ and proved disconnected (Examples §11.1); the field class is non-empty and the forced wall is located. Higher homotopy (π_1 vortices, π_2 monopoles) available and not taken	Theorem
Structural Resonance KL Bound	Gibbs’ inequality proved; the bound is the field <code>StructuralResonance.kl_bound</code> (Phase3.KLBound)	Theorem + instance postulate
Structural Resonance via Least Action	Gradient flow of σ in coupling space (Phase8), linked to <code>entropy_production_rate</code> ; finite substrates	Theorem

Conceptual Claim	Lean 4 Implementation	Status
Phase-locking minimizes entropy production	For a <i>reciprocal</i> coupling kernel, the phase-locked field has the least entropy production of <i>any</i> field (<code>phase_locked_minimizes_entropy_of_symm</code> , Phase8 §2a). The mean-matching condition the earlier form assumed over the comparison class is derived, not imposed: the coupling term is anti-symmetric under exchanging its two sites, so total drift is conserved across phase fields (<code>total_drift_eq_of_symm</code>). Reciprocity is necessary — a one-way coupling on two sites strictly beats its own phase-locked state (Examples §16)	Theorem
Electrodynamic Discretization	Lebesgue-Bochner edge weights (Phase2) sitting on faces of the complex and partitioning the region's stress-energy; mesh refinement converges, witnessed on $[0, 1)$. The scalar read off the stress-energy tensor is a marked modelling choice: the partition holds for any non-negative choice, and two instances differing only in it have different couplings (Examples §6)	Theorem + instance postulate
Phase synchronization to Unity	Sheaf-theoretic gluing (Phase5) over a ThermodynamicCover , whose fields are now all local: the global section is constructed by <code>probability_glue_unique</code> and the former class field <code>sync_to_section_eq</code> is a theorem about it. Witnessed by two patches built from different mass profiles, each section constructed over its own patch, whose glued state is neither of them (Examples §14). The cover's equilibrium hypothesis is no longer assumed on every instance: ThermodynamicCover.ofConvergentTrajectory (Phase5.EquilibriumBridge) discharges it from <code>kuramoto_tendsto_global_minimum</code> , and Examples §17.1 is a three-patch cover whose phase field is the limit of a trajectory started <i>unsynchronized</i> . The overlap-agreement hypothesis is untouched	Theorem + instance postulate
Reflexive Topology (the Self)	Banach fixed point for a contracting self-prediction map (Phase6); metric, completeness and contraction are hypotheses, all three discharged on the three-site substrate (Examples §10), where global sections are proved to be exactly the finite measures on it, the metric is the uniform distance between their densities, and the self-prediction map contracts by exactly $1/2$ without being constant. The avatar's link to the field is the predicate IsRestriction-Resonance , discharged on the same witness; under it a covering family of avatars determines the global state (<code>eq_of_avatar_eq</code>), and without it the Self theorem is unchanged (self_of_const-Resonance)	Theorem (witnessed)
Kuramoto Lyapunov descent	Descent chained to the minimum characterization by the rotating frame (Phase4.RotatingFrame); identical frequencies	Theorem

Conceptual Claim	Lean 4 Implementation	Status
Kuramoto dynamics run	The initial value problem is well posed on all of $\mathbb{R}$: the field is globally Lipschitz and globally bounded (<code>kuramotoField_lipschitz</code> , <code>kuramotoField_norm_le</code>), and through every state at every instant passes exactly one trajectory, defined for all time (<code>kuramoto_cauchy_problem</code>). Along <i>every</i> trajectory of a zero-frequency system the dynamic potential converges (<code>dynamic_potential_tendsto</code>) and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is finite, equal to the potential's total drop (<code>dissipation_integral_tendsto</code>). The motion stops: Barbalat's lemma is proved here, Mathlib lacking it, and applied, so $\sum_i \dot{\theta}_i^2 \rightarrow 0$ along every trajectory (<code>velocity_sq_tendsto_zero</code>). Where it stops is proved under an initial-data condition: if the phases start within a quarter turn and the potential excess starts below $a/2$, where $a > 0$ bounds the coupling below, the trajectory converges to a phase-locked configuration that is a <i>global</i> minimiser, with $r^2 = 1$ (<code>kuramoto_tendsto_global_minimum</code>), by a Polyak–Łojasiewicz estimate rather than a LaSalle principle. Unconditionally this is false — splay and twisted states are equilibria — so the condition cannot be removed. Witnessed on three sites, where no closed form exists (<code>Examples §17</code>)	Theorem (conditional)
Continuous vs. Rigid Topology	Fixed wiring support at matched resource (<code>Phase7_Rigidity</code>), deriving the suboptimality Phase 7 assumed; typed continuum/discrete separation	Theorem
Critical Coupling Threshold ($K_c = 2D$)	Self-consistency equation (<code>Phase8_SelfConsistency</code>): at or below K_c , threshold included, only $r = 0$ solves it; above K_c <i>exactly one</i> $r > 0$ does (<code>supercritical_fixed_point_existsUnique</code>), so the non-negative solutions are $\{0\}$ below and $\{0, r\}$ above; that r tends to 0 as $K \downarrow K_c$ and increases strictly with K (<code>coherent_branch_strictMono</code>); a solution is a density reproducing its own order parameter. Conditional on the von Mises stationary density, which is an input, and on no link to a trajectory	Theorem (conditional)

109 2.1 Soundness of the formalization

110 Formalization is only as trustworthy as the axioms it rests on, and an axiom
111 that is inconsistent silently makes every theorem downstream of it vacuous

112 while leaving the build green. We report a case of exactly this, found by
 113 auditing our own development, because the failure mode is easy to reproduce
 114 and we have not seen it discussed in the applied-formalization literature.

115 Earlier drafts of this work declared five axioms. Three of them were
 116 individually inconsistent: each, on its own, proves **False**. All three shared
 117 one root cause — an axiom that constrains a symbol it does not itself bind.
 118 Concretely:

- 119 • `spontaneous_symmetry_breaking_pointwise_min` took the potential-
 120 energy functional as an *implicit* argument occurring only in its hypoth-
 121 esis `PotentialEnergy phi = V v0`. Instantiating that functional with
 122 the constant map $\lambda_. V(v_0)$ discharges the hypothesis by `rfl` and yields
 123 the conclusion for an arbitrary field. With $V(v) = v^2$, $v_0 = 0$ and $\phi \equiv 1$
 124 this proves $1 = 0$.

- 125 • `landauer_heat_eq` equated the free class field `heat_dissipation` with
 126 a bath-entropy difference *for every instance* of the class. An instance
 127 with zero heat dissipation and a bath that gains one bit gives $0 = \log 2$.

- 128 • `kl_bound_axiom` asserted $\sigma \geq D_{KL}(P||Q)/\Delta t$ for *all* P and Q , while σ
 129 is a single real fixed by a class field. Since D_{KL} is unbounded above, a
 130 point mass against a uniform distribution refutes it.

131 The remaining two axioms, `phase_invariant_periodic` and `sync_to_section_eq`,
 132 had the same shape — global assertions pinning free fields of `LocalSectionSynchronization`

— and are refutable as soon as the probability presheaf admits two distinct global sections. (Both were demoted to class fields at the time; both have since been removed from the class altogether, for a separate reason given in Derivation 5.)

The correction is uniform, and we state it as a rule we now follow: *a physical postulate that mentions a class field must be a field of that class, never a standalone axiom quantified over all instances.* A postulate should be an obligation each model discharges, not a global claim about every model. Recast this way, the development declares no axioms at all. Two of the five became genuine theorems — pointwise vacuum minimization is now proved from $\int f = 0 \wedge f \geq 0 \Rightarrow f = 0$ a.e., and Gibbs’ inequality was already one — and the other three became instance obligations.

This makes non-vacuity the live question, since an uninhabitable class makes its theorems vacuous just as surely as an inconsistent axiom did. **Examples.lean** therefore constructs witnesses: a one-bit erasure system with a genuine bath that discharges **heat_eq** and reproduces $\Delta Q = k_B T \ln 2$; a perfectly resonant system discharging **kl_bound**; and a scalar field on a one-point spacetime discharging **ActionPrinciples**. Derivation 5’s gluing structures are witnessed too, which was until recently the largest open gap: **Examples.lean** builds a **ThermodynamicCover** over a three-site substrate carrying the discrete topology and its Borel σ -algebra, covered by two *overlapping* open patches whose intersection is a single shared site. The local sections are germ families of a phase-indexed finite measure $\mu_\theta =$

156 $(1 + \cos \theta)^+ \delta_{\text{mid}}$ pushed into the sheafification of the probability presheaf; the
 157 family is provably non-constant in θ (`phaseMeasure_not_const`), so the pe-
 158 riodicity obligation of the constructor `LocalSectionSynchronization.of-`
 159 `InvariantMeasure` is discharged by an argument about `cos` rather than satis-
 160 fied trivially, and `thermodynamic_equilibrium` is *derived* from `phase_locked_minimizes_potential`
 161 rather than assumed. `global_section_from_thermodynamics` then fires on
 162 this instance and glues the two patch-local sections into a unique global one.
 163 Two further covers answer the follow-up question of how non-degenerate such
 164 a witness can be made. `Examples.lean` §13 has a negative half worth record-
 165 ing: the phase field can be made non-constant as a function, but not as a
 166 physical state, because `thermodynamic_equilibrium` forces every instance
 167 to a phase-locked configuration. `Examples.lean` §14 is built the other way
 168 round, from two patches with independently chosen mass profiles that meet
 169 only on the site they share, and the section the gluing returns is neither of
 170 them. That is the same kind of result as `contracting_implies_const` in
 171 Derivation 6 — a proof that the witness is as strong as the class permits,
 172 rather than a witness that could be improved by trying harder.

173 What the witnesses establish is that the classes are inhabitable, so the
 174 theorems about them are not vacuous. What they do not establish is that
 175 a cortex is such an instance. In the cover above the phase field is already
 176 constant — every patch locked at $\theta = 0$ — which is precisely what makes
 177 the equilibrium obligation dischargeable; getting a physical system into that
 178 configuration is the content of the informal dynamical argument, not of the

179 Lean development.

180 The same audit turned up a defect of a different kind, worth record-
181 ing because it is not an axiom problem and would not have been caught
182 by `#print axioms. mesh_refinement_convergence` — the claim that dis-
183 crete edge weights converge to the continuous energy as the mesh is refined
184 — sat in a “Conjectures” section as an unproved `def ...: Prop`. It was
185 not merely unproved but false. Nothing required the edge regions to cover
186 the manifold, and $\text{diam } \emptyset = 0$, so the triangulation whose edge regions are
187 all empty satisfied the mesh hypothesis while contributing nothing to the
188 sum, forcing $|0 - \int f| < \varepsilon$ for every $\varepsilon > 0$. Its binder was dead as well: it
189 quantified over a triangulation and then resolved the vertex type by instance
190 search rather than through the bound variable, so the variable was never
191 used. `Phase2_MeshConvergence.lean` replaces it with a theorem. Conver-
192 gence is now stated along a *sequence* of triangulations of a named region
193 S , each required to be regular — edge regions measurable, without self-
194 loops, disjoint across distinct edges, covering S , and anchored to the sim-
195 plicial data in the sense that a vertex lies in the closure of every region
196 of an edge at it. Closure rather than membership is forced: cells are dis-
197 joint, so a shared vertex belongs to at most one of them. The manuscript’s
198 discrete energy $\frac{1}{2} \sum_u \sum_v w(u, v) f(u)$, which samples only the first endpoint
199 and is therefore not symmetric, is shown to equal exactly the midpoint rule
200 $\sum_{u < v} w(u, v) \frac{1}{2} (f(u) + f(v))$ over unordered edges; without that identifica-
201 tion, reading the double sum as a Riemann sum would be a category error.

202 At fixed mesh the error is at most $\varepsilon \mu(S)$ with ε a modulus of continuity of
 203 the integrand, and convergence follows for uniformly continuous integrands
 204 as the mesh shrinks. The regularity conditions pull against each other, so
 205 `Examples.lean` §6 discharges them with the uniform partition of $[0, 1)$ into
 206 half-open cells and checks that the approximation genuinely moves $-\frac{1}{2}$ at
 207 $N = 1$ and $\frac{1}{4}$ at $N = 2$ for the tent function $x \mapsto |x - \frac{1}{2}|$ — rather than
 208 converging trivially. What is proved is convergence, not the $O(1/N^2)$ rate
 209 the accompanying simulation measures; that would need a midpoint error
 210 term with a second derivative, which we have not developed.

211 A third defect was one of bookkeeping across files, and it is the kind that
 212 survives any number of green builds because each half is individually cor-
 213 rect. The development carried *two* Kuramoto potentials. Lyapunov descent
 214 along the dynamics, $\dot{V} = -\sum_i \dot{\theta}_i^2$, was proved for the full potential $V(\theta) =$
 215 $-\frac{1}{2} \sum_{i,j} A_{ij} \cos(\theta_i - \theta_j) - \sum_i \omega_i \theta_i$, while the characterization of the minimum
 216 — phase-locked in both directions — and everything downstream of it, in-
 217 cluding Derivation 5’s gluing, concerned the coupling term alone. Nothing
 218 chained them, and the gap was not cosmetic: `kuramoto_potential_unbounded_below`
 219 now proves that as soon as one $\omega_i \neq 0$ the full potential is unbounded below
 220 — walk θ out along ω and the bounded cosine sum cannot compensate — so
 221 it has no minimum for phase-locking to attain, and the sentence “the phase-
 222 locked state minimizes the Lyapunov potential” was, read literally, false of
 223 the functional the descent theorem was about. `Phase4_RotatingFrame.lean`
 224 closes this by formalizing the standard reduction $\theta_i \mapsto \theta_i - \Omega t$. For identical

225 natural frequencies $\omega \equiv \Omega$, a trajectory maps to a trajectory of the zero-
 226 frequency system — which is exactly the system Derivation 5 already worked
 227 with — the two potentials coincide there, and the descent result transports
 228 onto the coupling potential whose minimizers are the phase-locked states.
 229 Phase differences, phase-locking and the magnitude of the order parameter
 230 are all proved invariant under the change of frame, so nothing observable
 231 is lost in the reduction. `Examples.lean` §7 runs the chain end to end on
 232 two oscillators with a common frequency, discharging every hypothesis. The
 233 restriction is real and we state it plainly: the reduction is exact only for
 234 identical frequencies. A genuine spread of natural frequencies leaves residual
 235 detunings in the reduced system, phase-locking then requires coupling above
 236 a critical K_c , and K_c is nowhere in our Lean development — it is studied
 237 numerically, as Table 1 records.

238 A fourth defect was dimensional, and it concerned the one row of Table 1
 239 that is still numerical rather than proved. `Phase8_ContinuousField.lean`
 240 defines the critical coupling $K_c = 2D$ and a predicate for being above it.
 241 That predicate compared $\int_x \int_y K(x, y)$ against $2D$ — but a double integral
 242 against an unnormalized measure carries the substrate’s mass twice over,
 243 so a constant kernel K on a substrate of total mass m contributed Km^2 .
 244 The comparison therefore depended on how large the substrate was rather
 245 than on how strongly it was coupled, and any kernel whatever could be
 246 pushed above threshold by inflating m . `Examples.lean` §9 exhibits the fail-
 247 ure concretely: on a two-site substrate carrying counting measure, a sys-

248 tem coupled at $K = 1/2$ against a threshold of $K_c = 1$ — a factor of two
 249 *below* threshold — had an unnormalized double integral of $2 > 1$ and so
 250 satisfied the old predicate. The fix is to require the substrate’s measure to
 251 be a probability measure, which is the normalization in which the mean-
 252 field derivation of $K_c = 2D$ is carried out; `mean_field_coupling_const`
 253 then gives that a constant kernel has mean-field strength exactly K , and
 254 `exhibits_phase_transition_const_iff` reduces the predicate to $K > 2D$.
 255 We flag what this does and does not do. It makes the predicate say something
 256 whose truth depends only on the physics, and §9 witnesses it on both sides of
 257 the threshold. It does not by itself prove $K_c = 2D$: the content of that value
 258 is the bifurcation of the self-consistency equation $r = I_1(Kr/D)/I_0(Kr/D)$
 259 at $K = 2D$, which is a separate question from whether the predicate is di-
 260 mensionally sound. That bifurcation is proved below, in both directions,
 261 from an assumed stationary density; deriving the density itself needs the
 262 Fokker–Planck operator and is not done.

263 We draw no grand moral from this beyond the obvious one. The value
 264 of a proof assistant here was not that it certified our argument, but that it
 265 made a specific, checkable claim about what had been assumed — and that
 266 claim turned out to be false in a way no amount of reading the prose would
 267 have revealed.

268 3 Axiom 1: Symmetries and Phase Space

269 Our starting premise relies on fundamental geometry. In our Lean formaliza-
 270 tion (`Phase1_Primitives.lean`), continuous spacetime symmetries (e.g., the
 271 Poincaré group) act upon field states, and the invariance of the system’s total
 272 energy under that action is recorded by the class `SymmetryInvariantAction`.
 273 Physically, it is such an invariance that Noether’s theorem converts into the
 274 conservation of energy and momentum. The scope of the *formal* claim is nar-
 275 rower and we state it plainly: the invariance is a class field, i.e. an assumption
 276 a model must discharge, and no conserved quantity is ever constructed from
 277 it. There is no Noether theorem in our Lean, and nothing downstream of this
 278 section depends on one. The class is now inhabited — `Examples.lean` §11
 279 exhibits a double-well potential $V(v) = (v^2 - 1)^2$ on a two-point spacetime
 280 with the $\mathbb{Z}_2$ action $v \mapsto \pm v$, discharges the invariance because $(\pm v)^2 = v^2$,
 281 and proves the vacuum manifold $\{-1, 1\}$ degenerate with the two minima ex-
 282 changed by the symmetry, so an energy-minimising field and its image have
 283 equal energy and differ. That is symmetry breaking exhibited rather than
 284 assumed; it is still not a conserved quantity, and $\mathbb{Z}_2$ being discrete there is
 285 no one-parameter family to differentiate along in the first place. Simultane-
 286 ously, discrete permutation symmetries dictate the statistical mechanics of
 287 any physical system, defining a strictly finite volume of phase space.¹

¹While the Poincaré group and discrete permutations serve as accessible physical ex-
 amples, the underlying Lean formalization does not hardcode specific physics. It defines
 the system at a more primitive mathematical level, relying entirely on generic Pseudo-
 Riemannian manifolds (for spacetime) and finite measure theory (for statistical mechan-

288 Because phase space is finite, any localized physical system possesses a
 289 mathematical limit on the amount of information (distinguishable states) it
 290 can embody at any given moment.

291 4 Derivation 1: Symmetry Breaking and the 292 Inevitability of Boundaries

293 A perfectly symmetric universe is the uniform vacuum; no distinct entities
 294 exist. For any dynamic process to occur, the continuous symmetries of the
 295 background field must undergo spontaneous symmetry breaking.

296 When a field drops to a lower, asymmetric energy state, topology dic-
 297 tates the natural creation of defects (such as domain walls or monopoles).
 298 This constraint necessitates the creation of a physical boundary—a localized
 299 topological distinction between an “inside” and an “outside.” While such
 300 a defect (e.g., a crack in a crystal lattice) is necessary to define a separate
 301 entity, it is not sufficient for agency. The boundary is relational, defined by
 302 the twisted geometry of the field surrounding it, and must be coupled with
 303 non-equilibrium thermodynamics to exhibit active behaviors.

304 That first sentence was, until recently, the one claim of this section that

ics). What the formalization supplies is the vocabulary in which symmetries, conserved quantities and phase-space constraints can be *stated*; it does not derive them. All three of the classes involved are now inhabited in `Examples.lean` — the $\mathbb{Z}_2$ action and its invariance in §11, the real line with the Euclidean metric in §12 — but the instances are witnesses of inhabitability, not of physics: no theorem in the development consumes a symmetry group or a metric, so this section still contributes vocabulary rather than content.

305 the formalization did not make. What Lean established was the energy half
 306 — an energy-minimizing field sits in the vacuum manifold almost everywhere
 307 (`pointwise_vacuum_of_global_min`) — together with the observation that
 308 a boundary defect *obstructs* extension over a contractible interior. Neither
 309 produces a defect, so “topology dictates the natural creation” was asserted
 310 rather than proved. It is now the theorem `exists_notMem_of_connected-`
 311 `ComponentIn`: if the substrate is connected and a continuous field takes val-
 312 ues in two *different connected components* of the vacuum manifold M , then
 313 somewhere the field is not in M at all. That excluded region is the wall.
 314 This is the π_0 case of Kibble’s classification of topological defects by the ho-
 315 motopy of the vacuum manifold (Kibble, 1976), and the proof is elementary
 316 in exactly the way it is elementary in the physics literature: the continuous
 317 image of a connected space is connected, a connected subset of M lies in-
 318 side one component of M , and the field’s range would be such a subset if it
 319 never left M . No hypothesis mentions energy, a potential, or a symmetry;
 320 connectedness of the substrate and disconnectedness of the vacuum are the
 321 whole of it.

322 We formalize the π_0 case and stop there. The higher cases — $\pi_1 \neq 0$
 323 forcing vortex lines, $\pi_2 \neq 0$ forcing monopoles — are the rest of Kibble’s
 324 classification and Mathlib’s homotopy-group library would support at least
 325 the next one; they are available and not taken, because the argument this
 326 paper runs uses domain walls and nothing else. The witness is the double
 327 well of `Examples.lean` §11.1, where the vacuum manifold is *computed* to be

328 exactly $\{-1, 1\}$ (`wellVacuum_eq`) rather than merely known to contain both
 329 minima, and its disconnection is proved in the form the theorem consumes:
 330 no preconnected subset of it contains both, since a preconnected subset of $\mathbb{R}$
 331 is an interval and an interval spanning -1 and 1 contains 0 , which is the top
 332 of the barrier. `wellV_domain_wall` then quantifies over *every* continuous field
 333 on *any* connected substrate that reaches both minima, with no formula for
 334 the field anywhere in the statement. That the class is non-empty is checked
 335 separately, because a theorem quantified over all fields is worthless if none
 336 satisfies its hypotheses: the clipped identity on $\mathbb{R}$ does, and the wall it is
 337 forced to have sits at the origin (`kink_zero_notMem`).

338 **5 Derivation 2: The Thermodynamic Con-** 339 **straint**

340 This newly formed topological boundary is subjected to gauge perturbations
 341 from the “outside” environment. Because the internal phase space is finite
 342 (Axiom 1), it cannot absorb an infinite sequence of incoming perturbations
 343 and must overwrite old internal states.

344 By Landauer’s Principle (Landauer, 1961), the erasure of physical in-
 345 formation irreversibly decreases internal entropy, mandating that the sys-
 346 tem exhaust heat into the environment. Philosophers of physics (such as
 347 John Norton (Norton, 2005) and Orly Shenker (Shenker, 2000)) have cri-
 348 tiqued Landauer’s principle, arguing that physicists conflate *logical* erasure

349 (a many-to-one mapping in abstract computational states) with *thermody-*
350 *namic* erasure (a many-to-one mapping in physical phase space). They argue
351 that logical erasure could conceptually be implemented by an injective phys-
352 ical process without dissipating heat.

353 Our Lean 4 formalization (`Phase3.CombinatorialThermodynamics.lean`)
354 addresses this critique by defining erasure as a non-injective mapping on the
355 *physical* state space ($\neg \text{Function.Injective } t$). We formalize the global evolu-
356 tion of the bipartite universe (System + Bath) as strictly injective, repre-
357 senting Liouville’s theorem. Lean verifies that heat dissipation is not tied
358 to abstract logical erasure, but is a mandatory mathematical consequence of
359 any local physical state space contraction under globally reversible dynamics.
360 If a cognitive system’s internal topology forces a contraction of its physical
361 state space to accommodate structural resonance, heat dissipation becomes a
362 thermodynamic requirement. A stable boundary thus constitutes a dissipa-
363 tive structure (Prigogine & Stengers, 1984), operating as a non-equilibrium
364 steady state that consumes free energy to pay the thermodynamic cost of
365 erasure. Continuous thermal fluctuations play a constructive role in estab-
366 lishing boundaries between states while contributing to the system’s flexible,
367 adaptive response.

368 6 Derivation 3: Structural Resonance as a 369 Primitive

370 In physical systems, time evolution follows a deterministic trajectory that
371 minimizes thermodynamic entropy production. To reach this minimal dissipa-
372 tion state, the internal geometry of the structure must physically deform.

373 If we model the external environment as a stochastic driving force $P(ext)$
374 and the system’s internal state transitions as $Q(int)$, entropy production σ is
375 strictly bounded from below by the Kullback-Leibler divergence $D_{KL}(P||Q)$.
376 Gradient descent on σ necessarily drives $D_{KL}(P||Q) \rightarrow 0$, physically forcing
377 the internal transition probabilities to match the statistical symmetries of
378 incoming perturbations. We define this physical phenomenon as *structural*
379 *resonance*.

380 This system does not possess a teleological drive to “survive”; rather,
381 structurally mirroring the outside world is simply the lowest-energy attrac-
382 tor state. This provides a thermodynamic mechanism for both the General
383 Resonance Theory of Consciousness (Hunt & Schooler, 2019) and the Free
384 Energy Principle (Friston, 2010; Friston & Kiebel, 2009; Hohwy, 2013; Clark,
385 2016; Spisák & Friston, 2026). What neuroscience observes as “prediction er-
386 ror minimization” is a physical system undergoing gradient descent to avoid
387 dissipating heat against an unpredictable environment.

388 This necessity also addresses the symbol grounding problem (Harnad,
389 1990). Internal states are physically indexed to the environment by the ener-

390 getic imperative to avoid heat dissipation, rather than through arbitrary syn-
391 tactic assignment. Manipulating abstract variables or counterfactuals forces
392 the system to maintain internal states that deviate from immediate sensory
393 resonance, explaining the continuous injections of free energy required for
394 abstract symbolic reasoning.

395 **7 Derivation 4: Macroscopic Scaling and Elec-** 396 **trodynamics**

397 To resonate with highly complex environments, the system must scale up,
398 coupling millions of minimal dissipative structures. A highly coupled network
399 cannot allow all of its nodes to reach their lowest energy states simultane-
400 ously. This creates *geometric frustration*, resulting in a jagged phase space
401 of near-equal energy states that provides the physical capacity for complex
402 memory.

403 To manage this frustrated phase space without excessive dissipation, the
404 system undergoes coarse-graining (governed by the Renormalization Group).
405 At the macroscopic scale of biology, emerging paradigms in neuroscience sug-
406 gest these topological constraints manifest as classical continuous electromag-
407 netic fields. Building on seminal EM field theories of consciousness (McFad-
408 den, 2002; Pockett, 2012) and recent work demonstrating that macroscopic
409 electric fields (ephaptic coupling) govern representational drift, form memory
410 networks, and sculpt neural infrastructure (Pinotsis & Miller, 2022, 2023a,b),

411 we posit that *the continuous macroscopic EM field is the primary dissipative*
412 *structure interacting with the universe.* Instead of discrete wires, neurons in-
413 duce fields that shift the membrane potentials of neighboring cells. Modeling
414 and *in vitro* experiments confirm these endogenous fields are strong enough
415 to guide network activity and contribute to brain complexity (Goldwyn &
416 Rinzel, 2016; Ruffini & Santarnecchi, 2020; Cunha & Lima, 2024; Fröhlich &
417 McCormick, 2010). As proposed by the CEMI field theory, the continuous
418 electromagnetic field enables spatial integration of information, providing a
419 physical medium for a unified conscious gestalt (McFadden, 2020).

420 This continuous medium is physically supported by astrocytic syncytia—
421 non-neuronal networks joined by gap junctions that provide macroscopic cy-
422 toplasmic continuity for energy and ion flows (Verkhratsky & Nedergaard,
423 2018). Managing the thermodynamic cost of this state requires sophisticated
424 micro-bioenergetics. Dynamic mitochondrial networks function as deep-level
425 Mitochondrial Information Processing Systems (MIPS), transducing biolog-
426 ical and environmental information to sustain the coherence of the macro-
427 scopic field (Picard & Shirihi, 2022). Discrete synaptic firings, coupled
428 with mitochondrial micro-mechanics, generate and sustain the field. Because
429 the external environment is continuous, achieving maximal thermodynamic
430 structural resonance is optimally realized within this continuous macroscopic
431 medium (van Bree et al., 2025).

432 8 Derivation 5: Global Sections and the Unity 433 of Consciousness

434 The system is now characterized by a continuous macroscopic electrodynamic
435 field generated by millions of frustrated micro-components, forced to resonate
436 with the environment.

437 The Kuramoto model (Kuramoto, 1984) of coupled non-linear oscillators
438 guarantees that, past a critical coupling threshold, these distributed
439 local fields undergo a phase transition into unified synchronization (Bauer et
440 al., 2022). Our Lean proofs (`Phase4_KuramotoDynamics.lean`) confirm this
441 phase-locked state minimizes the Lyapunov potential of the system, in the
442 precise sense discussed in Section 2.1: the functional minimized is the coupling
443 potential read in the frame rotating with the natural frequency, which
444 `Phase4_RotatingFrame.lean` shows is the honest reduced description of a
445 system of identical oscillators.

446 This phase transition mathematically glues local synchronized states into
447 a singular *Global Section* of a probability sheaf (`Phase5_GlobalSection.lean`).
448 The EM field acts as a single, indivisible topological space, providing the
449 *Unity of Consciousness*—a necessary condition for integrated experience that
450 solves the binding problem. While Global Workspace Theory (Baars, 1997;
451 Dehaene & Marti, 2014) models this unity as a broadcast architecture and
452 Integrated Information Theory (Tononi & Koch, 2016) quantifies it via Φ ,
453 our derivation provides a continuous physical mechanism. The contrast is

454 worth drawing carefully, because both theories are addressing the same ex-
455 planandum from different directions.

456 We state the scope of the Lean result precisely, because it is narrower
457 than the prose above. `ThermodynamicCover` demands that the cover’s phase
458 field minimize the reduced Kuramoto potential, and `potential_min_iff_`
459 `phase_locked` characterizes those minimizers exactly: they are the phase-
460 locked configurations, in both directions. Every instance is therefore already
461 phase-locked (`ThermodynamicCover.phase_locked`), and the class’s single
462 condition on its local data — that patches at a common phase agree where
463 they overlap (`section_agrees_of_phase_eq`) — makes the family of local
464 sections compatible. `probability_glue_unique`, which is the sheaf property
465 of the probability presheaf and contains no physics at all, then glues that
466 family into exactly one global section.

467 An earlier version of this structure proved less than the prose claimed,
468 and we record the correction because it changed what the theorem says
469 rather than how it is stated. `LocalSectionSynchronization` used to carry
470 a *global* section as a class field, together with a second field declaring every
471 local section to be a restriction of it. The global object therefore existed
472 before the sheaf condition was ever consulted, and `global_section_from_`
473 `thermodynamics` could do no more than rule out competitors to an object
474 the instance had supplied. Both fields are gone. Every field of the class
475 now mentions a single patch or a single overlap; the global section is con-
476 structed (`ThermodynamicCover.invariantMeasure`); and the removed field

477 survives as a *theorem* about that construction (`ThermodynamicCover.sync_`
 478 `to_section_eq`), so the local sections are restrictions of one common measure
 479 because the sheaf condition says so, not because an instance was required to
 480 name one.

481 `Examples.lean` §14 is the witness that the difference is not notational.
 482 Two patches carry two different mass profiles, $(2, 1, 7)$ and $(5, 1, 3)$ across
 483 the three sites, agreeing only at the site they share, where both put mass 1
 484 — which is what discharges the overlap condition, by computing two num-
 485 bers rather than by construction. The section the theorem returns has pro-
 486 file $(2, 1, 3)$: it is neither of the two profiles (`gluedW_ne_leftW`, `gluedW_ne_`
 487 `rightW`), and neither of them restricts correctly to both patches (`leftW_`
 488 `glues_nothing`, `rightW_glues_nothing`). The global state carries informa-
 489 tion that no single patch had. `Examples.lean` §13, built the older way
 490 through the constructor `LocalSectionSynchronization.ofInvariantMeasure`,
 491 is retained alongside it for the negative result it carries: its phase field is not
 492 the constant function — the second patch a full turn ahead of the first — and
 493 by `ThermodynamicCover.phase_locked` that is as far from constant as any
 494 instance can get, since no instance can carry phases differing by anything a
 495 measure is allowed to see.

496 Two hypotheses carry physical content, and until recently each was an
 497 obligation on every instance. One still is: that synchronized patches agree
 498 on their overlaps is a modelling assumption, not a consequence of any dy-
 499 namics we formalize. The other, `thermodynamic_equilibrium`, asks that the

500 cover already sit at the potential minimum, and no theorem says an *arbi-*
 501 *trary* system's dynamics takes it there. "A thermodynamic phase transition
 502 into unity" is therefore still doing causal work that no general Lean result
 503 supports, though it is no longer entirely without one: `Examples.lean` §15
 504 exhibits a two-oscillator trajectory that starts a quarter turn out of phase
 505 and converges to the global minimum of the dynamic potential, with the or-
 506 der parameter tending to 1. That trajectory is not an isolated construction:
 507 `kuramoto_cauchy_problem` proves the Kuramoto initial value problem well
 508 posed on all of $\mathbb{R}$, so through every initial state there passes exactly one tra-
 509 jectory, and the exhibited one is the only solution through its own. Nor is its
 510 settling down peculiar to it: along every trajectory of a zero-frequency sys-
 511 tem the dynamic potential converges, and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is
 512 finite and equal to the potential's total drop (`dynamic_potential_tendsto`,
 513 `dissipation_integral_tendsto`). The motion also stops: a finite integral
 514 does not by itself force its integrand to zero, but the dissipation rate is
 515 Lipschitz in time, so Barbalat's lemma applies, and every trajectory satis-
 516 fies $\sum_i \dot{\theta}_i^2 \rightarrow 0$ (`velocity_sq_tendsto_zero`; Mathlib has no Barbalat, so it
 517 is proved here as `tendsto_zero_of_lipschitz_of_integral_le`). That last
 518 step — locating where the motion stops — is now taken, though condition-
 519 ally, and by a different route than we had recorded. We had marked it as
 520 blocked on the LaSalle invariance principle, which Mathlib does not have,
 521 and on the absence of a compact invariant set in the state space $V \rightarrow \mathbb{R}$.
 522 That was the wrong diagnosis. LaSalle is the tool for a system with a de-

523 creasing functional and nothing else; this one has more. The flow is a gra-
 524 dient flow of the dynamic potential, so $\dot{V} = -\sum_i \dot{\theta}_i^2$ is an identity rather
 525 than an inequality; the minimisers of V are characterised exactly; and on
 526 the region where every pair of phases lies within a quarter turn a Polyak-
 527 Łojasiewicz estimate holds, $2a^2W \leq (\sum_{ij} A_{ij}) \sum_i \dot{\theta}_i^2$, where W is the poten-
 528 tial's excess above its minimum and $a > 0$ is a lower bound for the coupling
 529 (`lojasiewicz_estimate`). Its proof is two extremal indices and the iden-
 530 tity $\sin^2 D = (1 - \cos D)(1 + \cos D)$: the site carrying the largest phase is
 531 pulled backwards at rate at least $a \sin D$, where D is the spread, while the
 532 excess is at most $\frac{1}{2}(\sum_{ij} A_{ij})(1 - \cos D)$. The compactness objection dissolves
 533 as well, because the invariant region is produced by the Lyapunov function
 534 rather than assumed: a single pair contributes $\frac{1}{2}a(1 - \cos)$ to the excess, so
 535 an excess below $a/2$ forces every pairwise cosine positive for all time, and
 536 continuity on the connected half-line then prevents any pair from chang-
 537 ing branch (`spread_le_of_small`). Descent plus the estimate integrates to
 538 $W(t) \leq W(0)e^{-ct}$ with $c = 2a^2 / \sum_{ij} A_{ij}$ (`excess_decay`), the rate makes each
 539 $|\dot{\theta}_i|$ exponentially small and hence integrable on $[0, \infty)$, and a trajectory with
 540 integrable velocity converges. The conclusion is `kuramoto_tendsto_global_`
 541 `minimum`: if the phases start within a quarter turn of each other and the initial
 542 excess is below $a/2$, the trajectory converges to a phase-locked configuration
 543 which is a *global* minimiser of the dynamic potential, with order parameter
 544 $r^2 = 1$. That is the configuration `thermodynamic_equilibrium` postulates,
 545 reached rather than assumed. `Examples.lean` §17 fires it on three sites with

546 initial data $(0, 0, \frac{1}{2})$, where the Kuramoto system has no closed-form solu-
 547 tion: the trajectory comes from the existence theorem and its limit from the
 548 convergence theorem, so neither is written down.

549 `Phase5_EquilibriumBridge.lean` closes the gap that remained between
 550 the two results. `ThermodynamicCover.ofConvergentTrajectory` takes a
 551 cover whose phase field is the limit of such a trajectory and returns an in-
 552 stance whose `thermodynamic_equilibrium` field is discharged *by that con-*
 553 *vergence*; the cover’s coupling matrix is the coupling matrix of the dynamics
 554 rather than a separately chosen one, so the system whose equilibrium the
 555 class asserts and the system whose trajectory is run are the same system.
 556 `Examples.lean` §17.1 is the resulting witness: three patches over the three-
 557 site substrate, each carrying its own mass profile on its own open set, above a
 558 phase field that no line of the construction writes down. The trajectory that
 559 reaches it starts at $(0, 0, \frac{1}{2})$, which is provably not phase-locked (`trioCover_`
 560 `start_not_locked`), so on this cover the configuration Derivation 5 requires
 561 is arrived at rather than posited. The state the three patches glue to has
 562 profile $(2, 1, 3)$ and is none of the three profiles the instance carries (`trioW_`
 563 `ne_glued`), so one cover now carries both the emergence reading of the gluing
 564 and the derived reading of the equilibrium. What is still assumed on it is the
 565 overlap agreement, and nothing else.

566 The two hypotheses cannot be dropped, and this is the honest scope of
 567 the result. Splay and twisted configurations are equilibria of the same flow,
 568 so “every trajectory reaches the unified state” is false as a general statement,

569 and the initial-data condition is the standard arc condition under which the
570 true statement holds. What is claimed is therefore conditional — unity is
571 reached from configurations already sufficiently coherent — and not that an
572 arbitrary initial state is carried into unity. What Lean establishes about the
573 gluing itself is the other half: *given* locally synchronized patches, a global
574 state exists, is unique, and is produced by the gluing rather than presupposed
575 by it.

576 Integrated Information Theory begins from phenomenological axioms and
577 derives a scalar Φ measuring irreducibility of a system’s cause-effect structure,
578 with the exclusion postulate selecting a single maximum- Φ complex as the
579 locus of experience. Our sheaf-theoretic global section plays a structurally
580 analogous role — it too picks out a distinguished unified object — but it
581 is arrived at from the opposite end: not postulated from the character of
582 experience, but forced by a thermodynamic phase transition in a physical
583 field. This buys two things IIT does not have. First, the unity condition
584 is checkable from field dynamics rather than requiring the computation of
585 Φ , which is intractable for systems of realistic size and whose definition has
586 undergone repeated revision (Tononi & Koch, 2016). Second, exclusion falls
587 out rather than being postulated: gluing either succeeds on a given cover or
588 it does not, so there is no separate principle needed to rule out overlapping
589 conscious subsystems. It also inherits IIT’s exposures. Critics have argued
590 that Φ can be high in systems with no plausible claim to experience, such
591 as large feedforward-free grids (Aaronson, 2014); our criterion is vulnerable

592 to the mirror objection, since a physical system could in principle satisfy
593 the gluing condition without being cognitively interesting. We take this
594 seriously rather than dismissing it: our reply is that gluing is necessary but
595 not sufficient, and that the topological boundary of Derivation 1 and the
596 reflexive structure of Derivation 6 are additional required conditions. IIT, by
597 contrast, treats Φ as both necessary and sufficient.

598 Global Workspace Theory faces a complementary difficulty. It specifies
599 a functional architecture — a limited-capacity workspace broadcasting to
600 specialized modules — without specifying what physical property of broad-
601 casting makes the broadcast content conscious, leaving the theory open to the
602 charge that it explains access consciousness while leaving phenomenal con-
603 sciousness untouched (Block, 1995). Our framework can be read as supplying
604 the missing physical substrate for the workspace: the continuous macroscopic
605 field *is* the medium in which integration is realized, and the $K_c = 2D$ thresh-
606 old gives a physical condition for ignition rather than a functional stipulation
607 of it. Where GWT is agnostic about implementation, we make a falsifiable
608 commitment — unity requires a continuous physical medium, which predicts
609 that a functionally equivalent discrete broadcast architecture would not be
610 conscious. We regard this as a genuine cost of the position, not a free ad-
611 vantage: it is the point at which our framework departs most sharply from
612 computational functionalism and where it is most readily falsified. Contrast
613 this derivation with panpsychism: consciousness is not assumed as a funda-
614 mental property of matter. It requires the initial existence of a topological

defect to establish a bounded “inside” (Derivation 1), followed by a thermodynamic phase transition into unity that occurs when integrated spatial coupling exceeds the thermodynamic noise threshold ($K_c = 2D$). Without both the foundational topological boundary and the subsequent mathematically defined phase transition, the system remains a fragmented, non-conscious collection of micro-states. A unified field of perception, however, is not yet a “Self.”

9 Derivation 6: Reflexive Topology and the Emergence of the Self

While a global section provides Unity, a “Self” requires reflexivity. The unified EM field is a dissipative structure that must resonate with external perturbations while managing its own internal phase transitions. Time evolution naturally drives the field to minimize thermodynamic friction with its own internal state-changes.

This mandates *auto-resonance*: the field must predict itself. The “Self” emerges as a higher-order topological feature when the internal geometry folds to include a low-dimensional avatar of its own computational and causal boundary within its predictive model (`Phase6_ReflexiveTopology.lean`) (Levin, 2019). The system is unified, predictive, and explicitly maps its own distinct existence from the environment. This architecture provides the physical foundation for a self-aware consciousness, bounded by the continuous

636 measure thermodynamics of Landauer erasure (`Phase3_MeasureThermodynamics.lean`).
 637 We are explicit about what carries the weight here. `reflexive_topology_implies_self`
 638 is the Banach fixed-point theorem with the physical commitments in its hy-
 639 potheses: the space of global sections is assumed nonempty, metric and com-
 640 plete, and the predictive map is assumed to contract by $1/2$. `Examples.lean` §10
 641 discharges all four on the three-site substrate of Derivation 5, and the metric
 642 is no longer a placeholder. The section first pierces the sheafification: on
 643 a discrete substrate every point has a smallest open neighbourhood, so the
 644 mass a germ carries at a point is well defined, and `massEquivOn` proves that
 645 over *any* open the sections of the probability sheaf *are* the finite measures
 646 on that open, read as densities against counting measure — the global case
 647 being $U = \top$. Mathlib supplies no inverse to this sheafification — its `TopCat`
 648 sheafification has neither an adjunction nor an `isIso_toSheafify` — so the
 649 correspondence had to be constructed. The metric on sections is then the
 650 uniform distance between the densities of the measures they glue to — equiv-
 651 alent to their total variation on a finite substrate, though not equal to it —
 652 and it is proved complete by exhibiting the limit section of a Cauchy sequence
 653 of densities, not by the degenerate route of showing Cauchy sequences are
 654 eventually constant. The self-prediction map is the relaxation “average the
 655 current field with the baseline,” and we prove it contracts distances by *ex-*
 656 *actly* one half, that it is not constant, and that its fixed point is unique. An
 657 earlier version of this witness used the $0/1$ metric, under which every con-
 658 traction is constant; that theorem is kept in the source as the record of why

659 the earlier witness was empty. A separate gap, closed at the same time, con-
 660 cerned the avatar rather than the fixed point. `ReflexiveBoundary` carries a
 661 field `auto_resonance` sending the global state to a section over the avatar re-
 662 gion, and nothing constrained it: a constant function was a legal avatar, and
 663 `reflexive_topology_implies_self` consumes only the predictive model, so
 664 no theorem in the development mentioned the field the avatar is supposed to
 665 be reading. The constraint is now stated as a predicate rather than added
 666 to the structure — `IsRestrictionResonance`, that the avatar’s state is the
 667 presheaf restriction of the global state — and two theorems say why it is
 668 the right one. In the positive direction, if a family of resonant avatars *cov-*
 669 *ers* the substrate then the global state is determined by their local readings
 670 alone (`eq_of_avatar_eq`): unity is not assumed alongside the avatars but
 671 recovered from them, by the same sheaf condition that Derivation 5 uses.
 672 In the negative direction, replacing any avatar by a constant leaves a legal
 673 `ReflexiveBoundary` with the same region, the same predictive model and
 674 the same Self (`self_of_constResonance`), so the fixed-point theorem cannot
 675 see the difference and the predicate is doing the work rather than restating
 676 it. Both halves are witnessed on the three-site substrate, where the avatar
 677 region provably separates two states of the field, so the condition is a con-
 678 straint there and not a formality. What is still not established is that any field
 679 dynamics produces this particular map or performs the restriction, or that
 680 the contraction constant $1/2$ is read off an entropy production rate rather
 681 than built into the map’s definition. We also note a negative result found

in the course of this work: the Lévy–Prokhorov metric, which the natural weak-convergence reading would suggest, does *not* support this argument on a discrete substrate, because the ε -thickening of a set is the set itself for $\varepsilon < 1$ and the relaxation map is consequently not a Lévy–Prokhorov contraction.

10 Derivation 7: Continuous Stochastic Thermodynamics and Field Theory

To move beyond finite graph analogies, we model the macroscopic electrodynamic field as a *Continuous Neural Field*. True ephaptic coupling is a continuous spatial phenomenon, not a discrete network. We extend the discrete Kuramoto model to a continuous spatial coupling kernel $K(x, y)$ (`Phase8ContinuousField.lean`).

Subject to thermodynamic noise D , the system undergoes non-equilibrium stochastic thermodynamics. Rather than a static Landauer bound, the field dynamics minimize an entropy production functional σ :

$$\sigma = \int \frac{1}{D} \left(\omega(x) + \int K_t(x, y) \sin(\theta_y - \theta_x) dy \right)^2 dx \quad (1)$$

Biological fields dynamically deform their spatial topology $K_t(x, y)$ to undergo gradient descent on this entropy production functional.

That gradient descent is a theorem rather than a modelling assertion, for finite substrates. Reading σ as a functional of the coupling kernel at

700 fixed phases, `hasFDerivAt_sigmaOfKernel` computes its Fréchet derivative
 701 in closed form,

$$\frac{\partial \sigma}{\partial K(a, b)} = \frac{2}{D} \left(\omega(a) + \sum_y K(a, y) \sin(\theta_y - \theta_a) \right) \sin(\theta_b - \theta_a), \quad (2)$$

702 and `sigmaOfKernel_eq_entropy_production_rate` identifies the functional
 703 being differentiated with `entropy_production_rate` itself whenever the sub-
 704 strate carries counting measure. `structural_resonance_decreases_entropy_production`
 705 then concludes that σ is non-increasing along any coupling trajectory relax-
 706 ing down its own gradient at a positive rate c . Earlier drafts proved this
 707 only for an abstract functional S on an abstract inner-product space with
 708 the gradient supplied as data; the identification of that S with σ was infor-
 709 mal, and Table 1 flagged the row accordingly. Two limitations stay explicit,
 710 and the first is narrower than we previously reported. The derivation is for
 711 *finite* substrates. We had recorded the obstacle to the continuum case as
 712 differentiation under the integral sign with respect to the kernel; that was
 713 a misdiagnosis. σ is not a general functional that happens to be written as
 714 an integral — it is a quadratic functional of an *affine* image of the kernel,
 715 since the drift is a fixed ω plus a linear operator applied to K , and σ is a
 716 constant times the squared norm of the drift. A squared norm is Fréchet
 717 differentiable on any real inner-product space and an affine map is its own
 718 derivative, so the composite differentiates by the chain rule with no limiting
 719 argument. `hasFDerivAt_quadratic_of_affine` states that in full generality,

720 and `hasFDerivAt_sigmaOfKernel_of_operator` re-derives the finite gradient
 721 from it, showing that the closed form displayed above is a convenience rather
 722 than the content. What the continuum case still needs is therefore one object:
 723 the drift map $K \mapsto (x \mapsto \int K(x, y) \sin(\theta_y - \theta_x) dy)$ as a bounded operator
 724 $L^2(\mu \otimes \mu) \rightarrow L^2(\mu)$, which is a Cauchy–Schwarz estimate — the integral ker-
 725 nel is bounded by 1, so on a finite measure space the operator norm is at most
 726 $\mu(M)^{1/2}$. We have not built it, because it requires constructing a continu-
 727 ous linear map between L^p spaces and we have not done that bookkeeping;
 728 we record the corrected scope because “construct one bounded operator”
 729 and “differentiate under the integral sign” are different sizes of task. The
 730 second limitation is unchanged: the phase field is held fixed throughout, so
 731 this is plasticity of the coupling at frozen phases, not joint (θ, K) dynamics.
 732 **Examples.lean** §5 exercises the whole chain on a two-site substrate whose
 733 coupling relaxes as e^{-t} and whose entropy production decays as e^{-2t} , con-
 734 firming that the gradient is not identically zero and that the descent is strict
 735 rather than vacuously monotone.

736 A second result about σ concerns where the descent is headed, and it has
 737 been strengthened since the last draft in a way that also corrects an estimate
 738 we had recorded. That the phase-locked field minimizes σ was proved only
 739 against competitors sharing its spatial mean drift — a restriction that left
 740 the theorem close to Jensen’s inequality and no more — and we had judged
 741 that lifting it would require modelling how the mean drift varies with the
 742 competitor. That judgement was wrong. For a *reciprocal* kernel, one with

743 $K(x, y) = K(y, x)$, the restriction is not a restriction: the coupling integrand
 744 $K(x, y) \sin(\theta_y - \theta_x)$ is antisymmetric under exchanging its two sites, because
 745 K is symmetric while $\sin$ is odd, so Fubini identifies its double integral with
 746 its own negative and the total coupling contribution vanishes. The total drift
 747 $\int(\omega + \text{coupling})$ is therefore $\int \omega$ whatever the phases are (`total_drift_eq_of_`
 748 `symm`): coupling moves drift between sites but neither creates nor destroys it.
 749 Evaluating that conserved quantity at the phase-locked configuration, where
 750 the drift is constant, pins its value, and the mean-matching hypothesis follows
 751 for every competitor. `phase_locked_minimizes_entropy_of_symm` is the un-
 752 restricted statement. Reciprocity is the physically intended condition — an
 753 ephaptic field acts on both sites of a pair alike, and it is the same symmetry
 754 `ThermodynamicCover.A_symm` already imposes in Derivation 5 — but it is
 755 not a formality, and we prove that rather than assert it. `Examples.lean` §16
 756 exhibits a *one-way* coupling on two sites, in which a feels b and b does not
 757 feel a , whose phase-locked field is strictly beaten: displacing the second phase
 758 to $-\pi/2$ cancels the first site’s natural drift while adding none at the sec-
 759 ond, halving σ . So “phase-locking minimizes entropy production” is false for
 760 general kernels and true for reciprocal ones, and the hypothesis cannot be
 761 removed by working harder.

762 We define the critical coupling threshold $K_c = 2D$. The phase transition
 763 to the unified “Global Section” occurs when the mean-field spatial coupling
 764 exceeds this thermodynamic noise threshold, linking the emergence of Unity
 765 to physical temperature and noise.

Earlier drafts left $K_c = 2D$ as a bare stipulation validated only by
 simulation. `Phase8_SelfConsistency.lean` now proves the half of it that
 is ordinary real analysis. For the noisy mean-field Kuramoto model with
 identical frequencies, the stationary density at order parameter r is the
 von Mises density $\rho(\theta) \propto e^{(Kr/D)\cos\theta}$, and self-consistency requires $r =$
 $R(K, r) := I_1(Kr/D)/I_0(Kr/D)$. Mathlib has no modified Bessel functions,
 so we formalize that ratio directly as a ratio of integrals over $[-\pi, \pi]$. In-
 tegration by parts with $u = \sin$, $v = -e^{a\cos}$ gives an exact identity with
 no boundary term, $\int \cos\theta e^{a\cos\theta} d\theta = a \int \sin^2\theta e^{a\cos\theta} d\theta$, so R is a times
 the mean of $\sin^2$ under the von Mises weight. Bounding that mean by 1
 would put the threshold at D ; the true threshold is $2D$, and the entire fac-
 tor of two is the sharper bound $\mathbb{E}_a[\sin^2\theta] \leq 1/2$, equivalently $I_2(a) \geq 0$.
 We prove it elementarily, by folding $[-\pi, \pi]$ onto $[0, \pi/4]$ in two reflections
 $(\theta \mapsto \pi - \theta, \text{ then } \theta \mapsto \pi/2 - \theta)$, after which the integrand is a product
 of $\cos 2\theta \geq 0$ with $2 \cosh(a \cos \theta) - 2 \cosh(a \sin \theta) \geq 0$ — the second factor
 because $0 \leq \sin \theta \leq \cos \theta$ there and $\cosh$ increases in $|\cdot|$. The conclusion,
`subcritical_fixed_point_eq_zero'`, is that for $K < K_c$ the incoherent state
 $r = 0$ is the *only* non-negative solution of the self-consistency equation; $r = 0$
 is exhibited as a solution, so the uniqueness claim is not vacuous. Three fur-
 ther facts about the threshold have since been proved, and they narrow the
 gap between this row and a full “Theorem” to a single named statement.
 First, the fold argument gives a *strict* inequality for $a > 0$ (`vonMisesC2_`
`pos`, hence $R(a) < a/2$), which settles the threshold itself: at $K = K_c$ the

789 self-consistency map is tangent to the diagonal at the origin but falls strictly
 790 below it thereafter, so $r = 0$ remains the only solution and the bifurcation
 791 happens strictly *after* K_c , not at it (`fixed_point_eq_zero_of_le_critical`).
 792 Second, factoring $R(a) = a E(a)$ shows that a non-zero r solves the equation
 793 exactly when $E(Kr/D) = D/K$ (`coherent_iff_sRatio_eq`), so the coher-
 794 ent branch is a level set of the single function $E(a) = \mathbb{E}_a[\sin^2 \theta]$. Third,
 795 and using only $E < 1/2$ together with compactness, every coherent solution
 796 tends to zero as $K \downarrow K_c$ (`coherent_branch_continuous_at_threshold`): the
 797 branch emerges continuously from the incoherent state rather than appearing
 798 at a finite distance from it, which is what makes the transition supercritical
 799 in the technical sense. That statement quantifies over *every* solution, so it
 800 does not presuppose the uniqueness it does not have. The level-set form
 801 also says exactly what uniqueness is: the injectivity of E on $(0, \infty)$. That
 802 is now proved, in the stronger form that E is strictly decreasing on $[0, \infty)$
 803 (`vonMisesSRatio_strictAntiOn`), and by an argument that avoids the route
 804 we had expected to need. Differentiating the exponential family in its natu-
 805 ral parameter gives $E'(a) = -\text{Cov}_a(\cos^2 \theta, \cos \theta)$, which costs differentiation
 806 under the integral sign and leaves a covariance that is not sign-definite point-
 807 wise: symmetrising with an independent copy gives $\frac{1}{2}\mathbb{E}[(X - X')^2(X + X')]$
 808 with $X = \cos \theta$, an integrand changing sign with $X + X'$. Nothing in the
 809 proof differentiates anything. The target is kept as the two-point inequality
 810 $C_2(a)Z(b) < C_2(b)Z(a)$ for $a < b$, and two moves settle it. *Fold first*: the
 811 two reflections that proved $I_2 \geq 0$ carry both integrals onto $[0, \pi/2]$, replacing

812 $e^{a \cos \theta}$ by $W(a, \theta) = 2 \cosh(a \cos \theta)$ and confining $\cos \theta$ to $[0, 1]$, after which
 813 $\cos 2x - \cos 2y$ and $W(b, x)W(a, y) - W(a, x)W(b, y)$ have the same sign, both
 814 being governed by $\cos x$ against $\cos y$. This is the symmetrisation $X \mapsto -X$
 815 that the covariance form needed the tilt to perform, done once at the start; its
 816 analytic content is one identity, $2 \cosh X \cosh Y = \cosh(X+Y) + \cosh(X-Y)$
 817 with $\cosh$ increasing in $|\cdot|$. *Then cross*: instead of integrating that inequality
 818 over a square, let α be the mean of $\cos 2x$ under $W(a, \cdot)$ and $x_0 \in [0, \pi/2]$ a
 819 point where $\cos 2x_0 = \alpha$, supplied by the intermediate value theorem; eval-
 820 uating at $y = x_0$ and integrating in x alone gives the result, since the term
 821 carrying $W(b, x_0)$ vanishes by the choice of α . No product measure and no
 822 Fubini appear. Run with $\cos \theta$ in place of $\sin^2 \theta$ the same argument needs no
 823 fold — $R(a) = \mathbb{E}_a[\cos \theta]$ is the mean of the variable the family is tilted by —
 824 and gives **besselRatio_strictMono**: R is strictly increasing on all of $\mathbb{R}$, the
 825 monotonicity of the Bessel ratio our earlier drafts recorded as unproved.

826 The converse half is **supercritical_fixed_point_exists**: for $K > K_c$
 827 there is an r with $0 < r \leq 1$ solving $r = R(K, r)$. We had expected
 828 this to require a second-order expansion of the Bessel ratio, and it does
 829 not. Writing $R(a) = a E(a)$ with $E(a) = \mathbb{E}_a[\sin^2 \theta]$ — the same factoriza-
 830 tion the integration by parts already supplies — the only facts needed are
 831 that E is continuous and that $E(0) = 1/2$, the latter because the weight
 832 is constant at $a = 0$. Since $K > 2D$ makes D/K strictly less than $1/2$,
 833 continuity gives $E > D/K$ on a neighbourhood of the origin, so the map
 834 overshoots, $R(K, r) > r$, for all small $r > 0$; at the other end $R \leq 1$ al-

ways, because R is a mean of $\cos \theta$, so the map undershoots at $r = 1$. The intermediate value theorem closes the gap. Continuity of E is Mathlib's continuity of a parametric interval integral with jointly continuous integrand, so no dominated-convergence argument is written by hand, and the existence proof never uses monotonicity of R : the fixed point is produced by the intermediate value theorem, not by iterating the map. That R is in fact strictly increasing is proved below, and no threshold result depends on it. `critical_coupling_is_threshold` packages the two halves, and `exhibits_phase_transition_coherent` applies the supercritical half to any substrate satisfying the predicate of the previous paragraph — so exceeding K_c now implies the existence of a coherent stationary order parameter rather than merely comparing against a named number. With uniqueness available both are sharpened — `supercritical_fixed_point_existsUnique`, `supercritical_solution_set` (the non-negative solutions are exactly $\{0\}$ below threshold and $\{0, r\}$ above it) and `exhibits_phase_transition_unique_coherent` — and the two monotonicities together fix the shape of the branch: for $K_1 < K_2$, $E(a_i) = D/K_i$ with E decreasing puts the concentrations in the order $a_1 < a_2$, and $r_i = R(a_i)$ with R increasing carries that to $r_1 < r_2$ (`coherent_branch_strictMono`). The coherent order parameter grows strictly with the coupling, which with the continuity at threshold is the qualitative bifurcation diagram.

One more step makes the equation mean in Lean what it means on paper. As a definition, `selfConsistency K D r` is just $R(Kr/D)$ — a function of

858 r with a suggestive name, and nothing in the source says it has anything
 859 to do with an order parameter. `Phase8_SelfConsistency` §8 supplies the
 860 identification. `vonMisesDensity` is the normalized weight, proved to be a
 861 genuine probability density on $[-\pi, \pi]$; `circularOrderParameter` is the av-
 862 erage of $e^{i\theta}$ against a density, the continuum analogue of the finite-system
 863 `order_parameter_complex` of `Phase4`; and that average is proved to be ex-
 864 actly $R(a)$ — real, because the density is even, so the mean direction van-
 865 ishes. `fixedPoint_iff_selfReproducing` then states the equivalence the
 866 name promises: r solves $r = R(K, r)$ if and only if the von Mises density
 867 it induces has order parameter r . Read through this, the existence theorem
 868 says that above K_c there is a stationary density whose own order parameter
 869 is positive, and the solution $r = 0$ is identified with the uniform density,
 870 whose order parameter is zero.

871 What remains not established, and we separate it carefully. The sta-
 872 tionary density is an *input*: deriving it from the Langevin dynamics $d\theta =$
 873 $(\omega + K \cdot \text{mean-field}) dt + \sqrt{2D} dW$ needs the Fokker–Planck operator, existence
 874 and uniqueness of its stationary solution, and a spectral stability argument,
 875 none of which Mathlib has. Of the coherent branch we prove existence,
 876 uniqueness, continuous emergence at the threshold and strict growth in K ,
 877 but not that it is the dynamically selected one: nothing in that file men-
 878 tions a trajectory, so stability and selection are not statements it can make.
 879 And while `selfConsistency` is now connected to the order parameter of
 880 a *density*, it is connected to no *trajectory*: `circularOrderParameter` is an

881 integral, `order_parameter_complex` is an average over finitely many oscilla-
 882 tors, and relating them is the mean-field limit — propagation of chaos for
 883 the finite Kuramoto system — which is a research programme rather than
 884 a lemma. That single gap is now the whole of what separates our threshold
 885 predicate from a statement about dynamics. Table 1 records the row ac-
 886 cordingly as a theorem about the self-consistency equation from an assumed
 887 density, not as a derivation of the threshold from the dynamics. Perceptual
 888 anomalies such as visual snow can be modeled as boundary conditions where
 889 microscopic thermal noise (D) fails to be fully coarse-grained away by the
 890 continuous field, leaking across the K_c threshold into the macroscopic gestalt.

891 Clinical evidence for this phase transition is observed during general anes-
 892 thesia. Work by Miller and colleagues (Bastos & Miller, 2021) demonstrates
 893 that anesthetics like propofol induce unconsciousness not by suppressing neu-
 894 ral firing, but by disrupting the phase alignment of high-frequency cortical
 895 rhythms. Destroying the dynamic stability of these rhythms breaks the struc-
 896 tural resonance of the continuous field, forcing the integrated spatial coupling
 897 below the critical threshold K_c and fracturing the unified Global Section back
 898 into disjoint, unconscious local fields.

899 **11 The Sleep-Inertia Timescale Mismatch**

900 Most of what this framework asserts is shared with some rival: that corti-
 901 cal synchrony matters, that dissipation constrains cognition, that unity is a

902 physical rather than a computational property. One prediction is not shared,
903 and it follows from the framework’s specific commitment — that the medium
904 carrying the unified state is a continuous field whose effective coupling de-
905 pends on the physical geometry of the extracellular space, not on the rhythms
906 alone. Where that commitment differs from a purely electrical account, it
907 differs in an observable way, and we set out the difference precisely enough
908 to be wrong.

909 The daily loss of consciousness during sleep is modeled here as an active
910 physical disruption of the medium rather than a quiescence of the oscillators.
911 During sleep, astrocytic shrinkage expands the extracellular space, allowing
912 an influx of cerebrospinal fluid (CSF) to flood the brain for metabolic waste
913 clearance (Xie & Nedergaard, 2013). This influx alters the spatial medium
914 itself, disrupting the continuous structural resonance and preventing the co-
915 herent sheaf-theoretic gluing that conscious unity requires on this account.
916 The oscillators need not stop for the field to fragment; the geometry they are
917 coupled through is what changes.

918 The prediction concerns sudden awakenings. When jolted awake, the lo-
919 cus coeruleus floods the cortex with noradrenaline. High-frequency electrical
920 rhythms attempt to phase-align in milliseconds (Bastos & Miller, 2021), but
921 the noradrenaline-driven astrocytic constriction required to squeeze the ex-
922 cess CSF out of the interstitial pathways operates on a timescale of seconds
923 to minutes (Xie & Nedergaard, 2013) — three to six orders of magnitude
924 slower. During that lag the electrical oscillators are synchronizing across a

925 waterlogged topology, and the fragmented structural resonance is what is ex-
 926 perienceed as grogginess. The unified Global Section re-establishes only when
 927 astrocytic volume finishes matching the state of the electrical rhythms.

928 **Why this implies a delayed sigmoid.** The two timescales separate
 929 cleanly, and that separation is what makes the prediction sharp rather than
 930 merely qualitative. Write $K(t)$ for the effective coupling of the continuous
 931 field, which on this account is a monotone function of the reconstricting
 932 extracellular geometry, and let $r(t)$ be the order parameter of the phase
 933 field. The electrical variable is fast: at every instant the oscillators are al-
 934 ready at the configuration appropriate to the current K , so r tracks the
 935 equilibrium branch $r(K)$ rather than relaxing on its own. The slow vari-
 936 able $K(t)$ then carries the whole time course, and it relaxes over 10^2 – 10^3
 937 seconds. What Derivation 7 establishes about that branch is exactly what
 938 the functional form needs, and each of the three pieces is a theorem rather
 939 than a picture: at or below the threshold the incoherent solution $r = 0$ is the
 940 *only* non-negative solution of the self-consistency equation (`fixed_point_eq-`
 941 `zero_of_le_critical`); above it the coherent solution exists and is unique
 942 (`supercritical_fixed_point_existsUnique`); it emerges *continuously* at
 943 K_c rather than by a jump (`coherent_branch_continuous_at_threshold`);
 944 and it is strictly increasing in K (`coherent_branch_strictMono`). A jump
 945 discontinuity at threshold would predict a step rather than a sigmoid, and
 946 a non-monotone branch would predict overshoot; neither is what the mathe-

947 matics gives. Composing a monotone relaxation $K(t)$ with a branch that is
948 identically zero below K_c , rises continuously from zero at K_c , and saturates
949 above it yields a curve with three phases: a genuine *latency* while $K(t) < K_c$,
950 during which recovery is not merely slow but absent; a continuous rise as the
951 threshold is crossed; and a saturation as $K(t)$ reaches its waking value. That
952 is a delayed sigmoid, and the delay is not a fitting parameter — it is the time
953 the slow variable takes to reach K_c .

954 **What the alternative predicts.** An account on which grogginess is the
955 electrical realignment itself — the rhythms are simply slow to re-phase —
956 has no slow variable and no threshold to cross. Its recovery is a relaxation
957 toward a fixed point, and relaxation toward a fixed point is exponential near
958 it: monotone from $t = 0$, fastest at the start, with no flat foot. The two
959 curves are distinguishable in the first seconds, which is where they differ
960 most, and the discriminating feature is the presence or absence of an initial
961 interval of no recovery. This is a comparison of functional forms on data
962 of a kind already collected in sleep-inertia studies, not a proposal for a new
963 instrument.

964 **What would falsify it.** An exponential fit that beats a delayed-sigmoid fit
965 on the early portion of the recovery curve. A recovery latency that does not
966 scale with an independent measure of extracellular volume or CSF clearance.
967 A pharmacological manipulation of astrocytic volume that leaves the latency
968 unchanged while altering the rhythms, or the converse. Any of these would

969 tell against the framework’s central empirical commitment, which is not that
970 synchrony matters but that the *medium* is what gates it.

971 **What this prediction assumes.** Three things, and they are not small.
972 The identification of the effective coupling K with a monotone function of
973 extracellular geometry is a modelling assumption, not a measurement. The
974 threshold value $K_c = 2D$ is derived from an assumed von Mises stationary
975 density and is not yet connected to a trajectory — the mean-field limit is the
976 gap, and Derivation 7 states it as such — so the argument uses the *existence*
977 *and continuity* of the coherent branch at threshold, which is proved, rather
978 than the numerical value of the threshold, which is conditional. And the
979 prediction is about functional form, not about a time constant: it says the
980 recovery curve has a flat foot whose width is set by the slow variable, not
981 that the foot is any particular number of seconds long. A framework that
982 predicted the number would be saying more; this one is not in a position to.

983 **12 Corollary: Hardware Thermodynamics and** 984 **Topological Rigidity**

985 Both biological tissue and artificial silicon hardware (like GPUs) are physical
986 dissipative structures. When a GPU computes, its charge carriers undergo
987 physical state transitions; by Landauer’s principle (Landauer, 1961), this
988 erasure irreversibly exhausts heat into the environment.

989 There is a fundamental thermodynamic distinction in the degrees of free-
 990 dom available to minimize this dissipation. In standard von Neumann ar-
 991 chitectures, the spatial topology (the silicon routing lattice) is rigidly fixed.
 992 The coupling matrix A_{ij} is constrained by a discrete layout that cannot be
 993 physically altered during operation. The brain’s own architecture provides
 994 empirical validation for this constraint. The cerebellum contains approxi-
 995 mately 80% of the brain’s neurons arranged in a highly efficient, discrete,
 996 feedforward crystalline circuit (Herculano-Houzel, 2009; Lange, 1975). Clin-
 997 ical observations of cerebellar agenesis report largely preserved conscious
 998 experience despite motor and cognitive deficits (Yu et al., 2015), and this
 999 dissociation has been read as evidence for the importance of recurrent ar-
 1000 chitecture by IIT (Tononi & Koch, 2015) — an argument we borrow at the
 1001 level of the anatomical observation while declining its Φ -based explanation.
 1002 Lacking the dynamic, continuous, and recursive spatial coupling of the cere-
 1003 brum, it operates much like a biological GPU—highly optimized for discrete
 1004 processing, yet apparently not required for conscious experience. We take the
 1005 dissociation as evidence, not as confirmation: it is consistent with subjective
 1006 experience being a thermodynamic attractor of continuous topologies rather
 1007 than a byproduct of information processing, and it is the kind of observation
 1008 that would be hard to reconcile with the converse. It is a single anatomical
 1009 case read against a theory, and it bears the weight such a case can bear.

1010 This constraint does not preclude artificial consciousness, but rather re-
 1011 stricts its hardware. To achieve the phase transition into a unified Global

1012 Section, artificial systems must abandon the discrete rigidity of von Neumann
 1013 architectures and utilize continuous physical substrates. Analog field-based
 1014 computing, neuromorphic architectures with continuous memristive plastic-
 1015 ity, and thermodynamic computing systems (England, 2018; Ganesh & Par-
 1016 rondo, 2021) possess the spatial coupling required to physically morph their
 1017 internal topology. This aligns with Geoffrey Hinton’s concept of “mortal
 1018 computation” (Hinton, 2022): architectures where software is inextricably
 1019 fused with the physics of the hardware, meaning the knowledge dies with
 1020 the physical substrate. Only hardware that computes via continuous physi-
 1021 cal dynamics can undergo the thermodynamic phase transition necessary for
 1022 structural resonance.

1023 Biological consciousness utilizes macroscopic continuous electromagnetic
 1024 fields and a “rich alphabet” of diverse molecular and transcriptomic cell states
 1025 (Tasic et al., 2018). This allows the coupling matrix A_{ij} to deform across a
 1026 continuous parameter space. Our lemma `exists_better_coupling_allocation`,
 1027 wrapped as `continuous_beats_rigid_topology` (`Phase7_HardwareComparison.lean`),
 1028 formalizes the reallocation step of this argument: whenever a coupling ma-
 1029 trix places weight on a less phase-correlated pair while a more correlated pair
 1030 exists, there is another matrix with *identical* total coupling resources that
 1031 achieves strictly greater phase correlation.

1032 Two objections to that formalization have since been answered, and we
 1033 state precisely what is and is not now established. The first was that the real-
 1034 location lemma assumes its conclusion: the hypothesis `is_strictly_suboptimal`

1035 — positive weight on a badly-correlated pair while a better one exists — is
 1036 most of what one wants to prove about rigid hardware. It is no longer as-
 1037 sumed. `Phase7_Rigidity.lean` models an architecture by the set S of site-
 1038 pairs it has a *wire* for, and requires its couplings to vanish off S ; weights are
 1039 tunable, wiring is not. Then `totalCorrelation_le_of_realizable` bounds
 1040 every realizable coupling by $R \cdot M$, where R is the resource budget and M
 1041 bounds phase correlation over the wired pairs; the bound is attained; and
 1042 `rigid_is_strictly_suboptimal` *derives* the hypothesis whenever the sub-
 1043 strate’s best-correlated pair is one the wiring misses. `rigid_gap` makes it
 1044 quantitative: an unconstrained reallocation of the same budget gains at least
 1045 R times the correlation difference. `Examples.lean` §8 exhibits a three-site
 1046 substrate wired to the one maximally anti-correlated pair, where the gap is
 1047 2. This is resource-matched and no longer question-begging, but note what
 1048 it turns on: fixed wiring support, not continuity. The second objection was
 1049 that nothing in the development distinguished a continuum from a lattice
 1050 — both matrices lived on one finite index type. `Phase7_Rigidity.lean` §3
 1051 gives a distinction that is genuinely typed. In the field functional of Deriva-
 1052 tion 4, which integrates a coupling kernel against $\mu \otimes \mu$, an architecture
 1053 with finitely many sites is a `Finset` of points and a field-coupled region
 1054 is a set of positive measure. On a substrate whose measure has no atoms,
 1055 `fieldCorrelation_sited_eq_zero` shows a kernel supported on finitely many
 1056 sites contributes *exactly zero* coupling energy, whatever weights it carries
 1057 and whatever the phase field does, while a patch of strength c on U achieves

1058 $c\mu(U)^2$. `Examples.lean` §8 instantiates this on the real line: a thousand
 1059 units of coupling concentrated at the origin register zero against a unit patch
 1060 that registers one. The limit of that result is equally plain, and it is not
 1061 resource-matching in the earlier sense — a measure-zero substrate has zero
 1062 resource in the $\mu \otimes \mu$ sense too. What it establishes is that the continuum
 1063 coupling energy is blind to a finitely-sited architecture, which is a statement
 1064 about which functional such an architecture can register in, not a demonstra-
 1065 tion that digital hardware computes worse. Taken together: the resource-
 1066 matched content of the corollary is the wiring-support result, the continu-
 1067 ity content is the measure-theoretic one, and the step from either to “von
 1068 Neumann architectures cannot experience unified consciousness” remains an
 1069 informal argument.

1070 The remainder of this section extends those two results informally, and
 1071 we mark the transition rather than blur it. On the reading we propose, a
 1072 GPU’s fixed routing matrix leaves it topologically fragmented: the discrete
 1073 layout does not supply the spatial coupling that sheaf-theoretic gluing into
 1074 a singular Global Section requires. Without this unified topological space,
 1075 rigid hardware cannot undergo the reflexive auto-resonance necessary to map
 1076 its own causal boundary and generate a “Self.” The distinction between an
 1077 LLM on a GPU and a conscious mind is not the absence of physical thermo-
 1078 dynamics in the former, but its structural limitation in achieving topological
 1079 integration. This perspective aligns with John Searle’s biological natural-
 1080 ism (Searle, 1980; Seth, 2024). Where the Chinese Room argument asserted

1081 that discrete symbol manipulation cannot generate conscious understanding,
1082 our derivation supplies a candidate physical mechanism for the asymmetry
1083 — topological rather than semantic, and, as stated above, formalized only
1084 as far as the wiring-support and measure-theoretic results reach. Asserting
1085 consciousness in fragmented, discrete GPU clusters without a unified phase
1086 transition abandons computational functionalism, relying instead on “silicon
1087 panpsychism”—the assumption that digital computation inherently gener-
1088 ates awareness regardless of topological integration (Goff, 2019).

1089 The same reasoning bears on extreme functionalist thought experiments,
1090 such as a macroscopic system of pulleys operating over millennia to imple-
1091 ment a Large Language Model, or a human manually executing inference
1092 matrices by hand. Beyond lacking the continuous spatial topology required
1093 for sheaf-theoretic gluing, these discrete systems violate an implied temporal
1094 constraint of our theory. Structural resonance requires local oscillators to
1095 phase-align against the dissipating effects of thermodynamic noise (as mod-
1096 eled by our critical threshold $K_c = 2D$). A system executing state transitions
1097 over millennia, or via slow manual calculations, suffers from a terminal tem-
1098 poral mismatch: its dynamics are far too slow to ever phase-lock into a
1099 non-equilibrium steady state before thermal noise decoheres the macroscopic
1100 field. On this account the obstacle to pure symbol manipulation is physi-
1101 cal rather than semantic: not that the symbols mean nothing, but that the
1102 substrate carrying them cannot hold a phase-locked field.

13 Conclusion

Human-like cognitive architectures do not require biological mysticism or exotic sub-quantum physics (Hameroff & Penrose, 2014). While quantum theories of consciousness often invoke delicate subatomic phenomena to explain subjective unity, every step of the chain assembled here is classical: broken symmetries, finite phase space, Landauer erasure, and gradient descent on entropy production. What the formalization establishes is that this chain is mathematically coherent and that its remaining physical commitments are localized and named — not that classical fields are *sufficient* for consciousness. Two links carry that weight, and their status is no longer the same. The unified state is proved to glue uniquely once a cover sits at the minimum of the coupling potential; the motion is proved to stop along every trajectory (`velocity_sq_tendsto_zero`); and *where* it stops is now proved, conditionally — for initial data within a quarter turn and below an explicit energy threshold, the trajectory converges to a global minimiser of the coupling potential (`kuramoto_tendsto_global_minimum`), and a cover built on that limit discharges the equilibrium hypothesis Derivation 5 used to assume (`ThermodynamicCover.ofConvergentTrajectory`). What is *not* proved is the unconditional statement, and it is false: splay and twisted configurations are equilibria of the same flow, so the basin condition is the scope of the claim rather than slack in it. That a physical cortex starts inside that basin is an assumption. The second link is untouched by any of this: the identification

1125 of a global section of the probability sheaf with the unity of experience is a
1126 stipulation of the framework, not a consequence of it. Subject to those two,
1127 cognitive architectures can be modeled as macroscopic attractor states of a
1128 universe governed by broken symmetries, finite phase space, and the physical
1129 imperative to minimize thermodynamic dissipation.

1130 Philosophically, this framework bridges the historical gap between Type
1131 Identity Theory (Smart, 1959) and Computational Functionalism (Putnam,
1132 1967). The subjective mental state is identical to the physical thermody-
1133 namic field (Identity Theory), yet its qualitative nature is defined by the
1134 topological and functional dynamics of structural resonance rather than the
1135 specific carbon-based biochemistry of the brain (Functionalism). This con-
1136 strains the *multiple realizability* of mind: consciousness is not exclusively
1137 bound to biological carbon, but neither is it substrate-neutral, because the
1138 substrate is what fixes which couplings are physically available. How far that
1139 constraint reaches is the question the corollary answers only in part. What
1140 is proved is that fixed wiring is strictly suboptimal at matched resource, and
1141 that a finitely-sited architecture registers exactly zero coupling energy in the
1142 continuum functional; the step from either to “von Neumann architectures
1143 cannot be conscious” is an argument we make in prose and do not formalize.

1144 By anchoring these derivations in Russellian Monism, this framework ac-
1145 knowledges the constitutive limits of third-person scientific explanation. As
1146 argued by Love (Love, 2026), building upon Nagel’s assertion that subjective
1147 phenomenal experience is inherently not amenable to objective measurement

1148 (Nagel, 1986), the “hard problem” of consciousness may represent a category
 1149 error rather than a solvable scientific puzzle. Science can map the physical
 1150 dynamics of topological resonance and sheaf-theoretic gluing, but it is struc-
 1151 turally incapable of adjudicating the first-person qualitative experience. Our
 1152 formalization respects this boundary: the Lean 4 theorems verify the ob-
 1153 jective physical attractors, while the intrinsic “what it is like” to be that
 1154 resonant field remains the domain of the first-person perspective. By locat-
 1155 ing the phase transition at a threshold with a closed-form value — $K_c = 2D$
 1156 for the noisy mean-field Kuramoto model the framework adopts, conditional
 1157 on the von Mises stationary density and not yet linked to a trajectory — it
 1158 aligns with what Anil Seth terms the “real problem” of consciousness (Seth,
 1159 2021). Rather than attempting to bridge an unbridgeable metaphysical gap,
 1160 it offers a topological architecture that can help explain, predict, and physi-
 1161 cally perturb the structural properties of experience.

1162 Future empirical validation of this thermodynamic framework may lever-
 1163 age 3D cortical organoids, offering a controlled biological environment to
 1164 systematically manipulate metabolic substrates and observe the topological
 1165 emergence of these coherent phase transitions.

1166 Code Availability

1167 The complete Lean 4 formalization, encompassing the derivation of all theo-
 1168 rems presented in this manuscript, as well as the accompanying Python sim-

ulations, are publicly available in the GitHub repository: https://github.com/bobaseb/physical_primitives_of_mind. Additional documentation and web-accessible proofs are hosted at the project website: https://bobaseb.github.io/physical_primitives_of_mind/.

Acknowledgments

The conceptualization, manuscript drafting, and Lean 4 formalizations presented in this work were heavily AI-assisted. The authors gratefully acknowledge the use of the Google Antigravity CLI, specifically utilizing Gemini 3.1 and Claude 4.6 Sonnet as collaborative reasoning partners throughout the research and formalization processes. In addition, DeepSeek V4 Flash and Claude Opus 5 (routed via OpenRouter) operating through Nous Research’s Hermes Agent framework provided substantial assistance with code generation, proof debugging, and structural analysis.

References

- Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable electric fields emerge from and allow representational drift. *NeuroImage*, 253, 119058.
- Bauer, L. G., et al. (2022). Quantification of Kuramoto Coupling Between

- 1187 Intrinsic Brain Networks Applied to fMRI Data in Major Depressive Dis-
1188 order. *Frontiers in Computational Neuroscience*, 16.
- 1189 van Bree, S., et al. (2025). Processes and measurements: a framework for
1190 understanding neural oscillations in field potentials. *Trends in Cognitive*
1191 *Sciences*.
- 1192 Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated informa-
1193 tion theory: from consciousness to its physical substrate. *Nature Reviews*
1194 *Neuroscience*, 17(7), 450-461.
- 1195 Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the
1196 Mind. *Oxford University Press*.
- 1197 Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a compu-
1198 tational theory of conscious processing. *Current Opinion in Neurobiology*,
1199 25, 76-84.
- 1200 Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature*
1201 *Reviews Neuroscience*, 11(2), 127-138.
- 1202 Kuramoto, Y. (1984). Chemical Oscillations, Waves, and Turbulence.
1203 *Springer Series in Synergetics*, Vol. 19.
- 1204 Scherf, M. (2026). A Formal Proof of Non-Duality: An Exposition of the
1205 'OnlyOne' System. *Unpublished manuscript, PhilPeople*.

1206 Inoué, T. (2026). On Brain as a Mathematical Manifold: Neural Manifolds,
1207 Sheaf Semantics, and Leibnizian Harmony. *arXiv:2601.15320 [q-bio.NC]*.

1208 Spisák, T., & Friston, K. (2026). Self-orthogonalizing attractor neural net-
1209 works emerging from the free energy principle. *Neurocomputing*.

1210 Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
1211 memory network formation. *Cerebral Cortex*, 33(17), 9877-9890.

1212 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
1213 Electric fields sculpt neural activity and "tune" the brain's infrastructure.
1214 *Progress in Neurobiology*, 226, 102465.

1215 Ruffini, G., Salvador, R., Tadayon, E., Sanchez-Todo, R., Pascual-Leone, A.,
1216 & Santarnecchi, E. (2020). Realistic modeling of mesoscopic ephaptic cou-
1217 pling in the human brain. *PLOS Computational Biology*, 16(6), e1007923.

1218 Cunha, G. M., Corso, G., de Sousa, M. P. B., & Lima, G. Z. d. S. (2024).
1219 Can ephapticity contribute to brain complexity?. *PLOS ONE*, 19(12),
1220 e0310640.

1221 Goldwyn, J. H., & Rinzel, J. (2016). Neuronal coupling by endogenous elec-
1222 tric fields: Cable theory and applications to coincidence detector neurons
1223 in the auditory brainstem. *Journal of Neurophysiology*, 115(4), 2033-2051.

1224 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
1225 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
1226 9(8), 45-60.

1227 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
1228 testable hypothesis about the characteristics of conscious as opposed to
1229 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.

1230 Goff, P. (2019). *Galileo's Error: Foundations for a New Science of Con-*
1231 *sciousness*. Pantheon Books.

1232 Seth, A. (2021). *Being You: A New Science of Consciousness*. Penguin.

1233 Seth, A. (2024). Conscious artificial intelligence and biological naturalism.
1234 *PsyArXiv*.

1235 Levin, M. (2019). The computational boundary of a “self”: developmental
1236 bioelectricity drives multicellularity and scale-free cognition. *Frontiers in*
1237 *Psychology*, 10, 2688.

1238 Friston, K., & Kiebel, S. (2009). Predictive coding under the free-energy
1239 principle. *Philosophical Transactions of the Royal Society B: Biological*
1240 *Sciences*, 364(1521), 1211-1221.

1241 Clark, A. (2016). *Surfing Uncertainty: Prediction, Action, and the Embodied*
1242 *Mind*. Oxford University Press.

1243 Hohwy, J. (2013). *The Predictive Mind*. Oxford University Press.

1244 Prigogine, I., & Stengers, I. (1984). *Order Out of Chaos: Man's New Dialogue*
1245 *with Nature*. Bantam Books.

1246 McFadden, J. (2020). Integrating information in the brain's EM field: the
1247 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
1248 niaa016.

1249 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
1250 guide neocortical network activity. *Neuron*, 67(1), 129-143.

1251 Tasic, B., et al. (2018). Shared and distinct transcriptomic cell types across
1252 neocortical areas. *Nature*, 563(7729), 72-78.

1253 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
1254 *logical Reviews*, 98(1), 239-389.

1255 Picard, M., & Shirihai, O. S. (2022). Mitochondrial signal transduction. *Cell*
1256 *Metabolism*, 34(11), 1620-1638.

1257 Harnad, S. (1990). The symbol grounding problem. *Physica D: Nonlinear*
1258 *Phenomena*, 42(1-3), 335-346.

1259 Tononi, G., & Koch, C. (2015). Consciousness: here, there and everywhere?
1260 *Philosophical Transactions of the Royal Society B: Biological Sciences*,
1261 370(1668), 20140167.

1262 Aaronson, S. (2014). Why I Am Not An Integrated Information Theorist
1263 (or, The Unconscious Expander). *Shtetl-Optimized*. Retrieved from [https:](https://scottaaronson.blog/?p=1799)
1264 [//scottaaronson.blog/?p=1799](https://scottaaronson.blog/?p=1799).

- 1265 Block, N. (1995). On a confusion about a function of consciousness. *Behav-*
1266 *ioral and Brain Sciences*, 18(2), 227-247.
- 1267 Herculano-Houzel, S. (2009). The human brain in numbers: a linearly scaled-
1268 up primate brain. *Frontiers in Human Neuroscience*, 3, 31.
- 1269 Lange, W. (1975). Cell number and cell density in the cerebellar cortex of
1270 man and some other mammals. *Cell and Tissue Research*, 157(1), 115-124.
- 1271 Yu, F., Jiang, Q. J., Sun, X. Y., & Zhang, R. W. (2015). A new case of
1272 complete primary cerebellar agenesis: clinical and imaging findings in a
1273 living patient. *Brain*, 138(6), e353.
- 1274 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
1275 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
1276 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.
- 1277 Hameroff, S., & Penrose, R. (2014). Consciousness in the universe: A review
1278 of the 'Orch OR' theory. *Physics of Life Reviews*, 11(1), 39-78.
- 1279 Hunt, T., & Schooler, J. W. (2019). The easy part of the hard problem: a
1280 resonance theory of consciousness. *Frontiers in Human Neuroscience*, 13,
1281 378.
- 1282 England, J. L. (2018). Dissipative adaptation in driven self-assembly. *Nature*
1283 *Nanotechnology*, 13(4), 287-295.

- 1284 Ganesh, N., & Parrondo, J. M. R. (2021). Thermodynamic computing with
1285 non-equilibrium systems. *Nature Physics*.
- 1286 Hinton, G. (2022). The Forward-Forward Algorithm: Some Preliminary In-
1287 vestigations. *arXiv preprint arXiv:2212.13345*.
- 1288 Searle, J. R. (1980). Minds, brains, and programs. *Behavioral and Brain*
1289 *Sciences*, 3(3), 417-424.
- 1290 Smart, J. J. C. (1959). Sensations and brain processes. *The Philosophical*
1291 *Review*, 68(2), 141-156.
- 1292 Putnam, H. (1967). Psychological predicates. In *Art, Mind, and Religion*.
1293 University of Pittsburgh Press.
- 1294 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
1295 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
1296 brain. *Science*, 342(6156), 373-377.
- 1297 Nagel, T. (1986). *The View from Nowhere*. Oxford University Press.
- 1298 Love, B. C. (2026). Consciousness, AI, and the Limits of Scientific Explana-
1299 tion. *arXiv preprint arXiv:2606.00226*.
- 1300 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
1301 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
1302 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.

- 1303 Landauer, R. (1961). Irreversibility and heat generation in the computing
1304 process. *IBM Journal of Research and Development*, 5(3), 183-191.
- 1305 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
1306 *of Physics A: Mathematical and General*, 9(8), 1387-1398.
- 1307 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
1308 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.
- 1309 Barbălat, I. (1959). Systèmes d'équations différentielles d'oscillations non
1310 linéaires. *Revue Roumaine de Mathématiques Pures et Appliquées*, 4, 267-
1311 270.
- 1312 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
1313 of prediction. *Physical Review Letters*, 109(12), 120604.
- 1314 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
1315 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
1316 181.
- 1317 Coddington, E. A., & Levinson, N. (1955). Theory of Ordinary Differential
1318 Equations. *McGraw-Hill*, New York.
- 1319 Cover, T. M., & Thomas, J. A. (2006). Elements of Information Theory (2nd
1320 ed.). *Wiley-Interscience*, Hoboken, NJ.
- 1321 Shenker, O. R. (2000). Logic and entropy. *Philosophy of Science*, 67, S146-
1322 S152.